

# Biology-Compliance Audit

Membrane / Formin / Arp2/3 Coarse-Grained Lamellipodium Model — BoxOfActin

Branch: `membrane-render-and-dense-v5` · Date: 2026-06-18 · Scope: formin-seeded mothers, Arp2/3 branching, capping/localization, membrane collision & yield.  
Method: static read of `Membrane.java`, `FilSegment.java`, `Arp23.java`, `StickyNode.java`, `Env.java`, `ActA.java`, `ProteinNode.java`, `NodeLink.java`; cross-checked against the DTS design note and the project journal.

## 0. The most important structural finding

There are two membrane models in the tree, and the audit request is written in the vocabulary of the legacy one. Every parameter named in the request exists — but the formin/Arp/capping/align/yield family belongs to the **legacy node-lattice membrane**, which **every DTS config explicitly disables** (`buildMembraneSheet:false`; `forminConc:true:0.0`; `arpConc:true:0.0`). The live production model is the **DTS (dynamically-triangulated surface)** family of `dts*` parameters. The legacy names are present, mostly runtime-mutable and documented, but **dead on the production path**.

|                            | Legacy "node-lattice" membrane                                                                                                                                        | Current DTS membrane (active)                                                                                                                 |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| Builder                    | <code>StickyNode.makeSheetHexPackedNodes</code> / <code>makeSphereOfNodes</code>                                                                                      | <code>Membrane.buildIcosphereMembrane</code>                                                                                                  |
| IC gate                    | <code>buildMembraneSheet</code> / <code>buildMembraneSphere</code> (default false)                                                                                    | <code>buildDtsMembrane</code> (true in all real configs)                                                                                      |
| Formin pool                | <code>forminConc</code> , <code>forminConsumePerMother</code> , <code>forminRecover</code> , <code>deNovoNucleate</code>                                              | <code>dtsForminPool/Consume/NucRate/Recover</code> , <code>Membrane.forminStep</code>                                                         |
| Arp field                  | <code>arpLocalField</code> , <code>arpDiffusion</code> , <code>arpBulkExchange</code> , <code>arpConsumePerBranch</code>                                              | <code>dtsArp0n/Target/Diffusion/Decay</code> , <code>Membrane.branchStep</code>                                                               |
| Zone / cap / align / yield | <code>branchMembraneDist</code> , <code>membraneCapDist</code> , <code>membraneAlignTorque/Angle</code> , <code>motherTetherDepth</code> , <code>membraneYield</code> | <code>dtsForminStandoff</code> , <code>dtsActinPushMax</code> , <code>ratchet</code> , <code>nodeTorqSpring</code> , native area/volume yield |

The rest of this audit evaluates the **live DTS path** and flags the legacy duplicates as a clarity hazard.

## 1. Matches biology MATCHES

- Arp2/3 never de-novo nucleates.** `Membrane.branchStep` only creates a daughter off an existing `FilSegment` mother whose barbed tip is within ~60 nm of the cortex and sits in a high local-Arp region. No path lets the Arp field alone spawn a filament. (**core requirement #2**)
- Formin is the sole mother source.** `forminStep` → `makeForminMother` builds an uncapped *linear* seed (no Arp junction) that becomes a branching template. Mother-generation (formin) and daughter-generation (Arp2/3) are cleanly separated.
- Branch angle = 70°** (`arp23AlphaAngle`), the canonical dendritic Arp2/3 angle, with helical-azimuth modulation so daughters fan around the mother.
- Barbed end held at the cortex.** With `dtsForminGrowOut=true` the mother is seeded barbed-end-out (`uVec = outward`), `end2` linked to the vertex; the pointed end trails into cytoplasm — the correct formin geometry.
- Mogilner–Oster ratchet.** `ratchetPolyFactor` throttles barbed-end elongation by `exp(-F·(6-gap)/kT)` using the tip → membrane clearance (`end2TipC`) written by the DTS collision. Growth stalls as load → stall force and resumes as the cortex yields — polymerization genuinely becomes protrusion.
- Continuous surface, not a porous lattice.** `segmentVsMembrane` samples the whole rod against tiled triangles (no face-center gaps) and distributes the push barycentrically to 3 vertices. The Stage-3-prep containment sweep passed in all orientations.
- Finite mother seeding.** `dtsForminPool=1`, `dtsForminConsume=1`, `dtsForminRecover=0` ⇒ ~1 mother per hot vertex, no recovery — a hard, finite cap. (**#3, #4**)
- Arp reaction–diffusion.** Hot patches are clamped sources; bulk decay (`dtsArpDecay`) + per-branch consumption (`dtsArpConsumePerBranch`) are the sinks; halo length ≈ √(D/k). Source-at-NPF / sink-in-bulk is correct. (**#10**)

## 2. Coarse but acceptable COARSE-OK

- Force-sensitive formin release** (`forminCanHold`): exponential catch under compression, reference force hard-coded at 2 pN. Plausible but uncalibrated; configs set `forminRelease:0` anyway.

- **Daughter network embedding** is approximated by a rotational-drag floor ( `filDragMinMonomers` ), critically-damped junctions ( `arpTransFracMove=1.0` ), and inherited Brownian damping ( `heldBrownFactor` ) rather than explicit geometry. Phenomenological but stable.
- **Held-mother thermal damping** ( `forminBarbedBrownianFactor` =0.05, `arpHeldBrownianFactor` =0.02): a held filament *should* have suppressed thermal; the 20×/50× factors are unprincipled but qualitatively right.
- **DTS "yield" is native, not the legacy `membraneYield` flag.** Actin push moves vertices through the ordinary overdamped integrator; bending+area+volume energies resist; `subcycleRelaxAll` is retired. This is *more* physical than the legacy re-apply-each-Jacobi-pass scheme. (#12)
- `dtsMaxFilaments` **hard cap** as a runaway backstop — blunt but harmless if consumption feedback is the real limiter.
- **Solid shell, not yet a fluid bilayer** — bond flips (design §4) are unimplemented, so there is no lateral lipid flow / area transport into a finger yet. Acceptable for protrusion tests; state it explicitly.

### 3. Potential mismatch MISMATCH

| ID | Issue                                                                      | Detail & impact                                                                                                                                                                                                                                                                                                                                                                            |
|----|----------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| P1 | Cortex-localized capping is <b>OFF</b> on the live path                    | <code>membraneCapDist=0</code> everywhere $\Rightarrow$ <code>checkCapping</code> never uses the membrane-distance rule; barbed ends cap only stochastically/sterically. <b>Nothing keeps growth a thin leading-edge layer.</b> <code>end2TipC</code> is already written by the DTS collision, so the rule would work if enabled. <i>The biggest biological gap; config-only fix.</i> (#8) |
| P2 | Mother orientation has no sustained normal bias on DTS                     | The clean mechanism ( <code>membraneAlignTorque</code> $\rightarrow$ normal, <code>membraneAlignAngle</code> ) is gated on <code>StickyNode.sphericalGeometry</code> — legacy-only. The DTS bond's <code>nodeTorqSpring</code> aligns to the vertex's <i>arbitrary body frame</i> at $\sim 1e-18$ N/rad $\approx$ off. DTS mothers are oriented only by seed geometry. (#6, #7)            |
| P3 | Legacy <code>membraneNodeDragScale</code> still defaults to the 1e-10 trap | DTS <code>MembraneVertex</code> bypasses it with physical drag, but enabling the legacy sheet/sphere path re-inherits the known force-swallowing bug. (#11)                                                                                                                                                                                                                                |
| P4 | De-novo nucleators that bypass Arp2/3 exist                                | <code>ActA.nucleateFilament</code> and generic <code>ProteinNode</code> nucleation. <b>Off in lamellipodium configs</b> (no current violation), but <code>ActA</code> is biologically wrong for <i>Listeria</i> (should recruit Arp2/3) and <code>ProteinNode</code> nucleation is test-only. Document, don't ship as dendritic biology. (#2, #13)                                         |
| P5 | Pre-existing DTS bending $1/\sin\theta$ blow-up                            | When actin drives a vertex into a near-degenerate triangle, the exact dihedral gradient $\rightarrow \sim 1e128$ N, bounded only by the <code>dtsMaxDispFrac</code> clamp. This — not the force caps — is the real numerical-stability risk for push $\rightarrow$ protrusion. (#12)                                                                                                       |
| P6 | Comment rot                                                                | <code>Membrane.forminStep:535–536</code> ("grow radially inward, barbed into the cytoplasm, pointed end at the inner face") describes the <i>legacy</i> branch but sits above both branches — it contradicts the active barbed-out geometry.                                                                                                                                               |

### 4. Answers to the 13 audit questions

- Separates formin mothers from Arp daughters?** Yes — distinct paths/flags ( `forminMother` vs Arp23 junction). ✓
- Arp de-novo nucleation contradicting biology?** Not on the dendritic path. `ActA` / `ProteinNode` de-novo exist but are off in lamellipodium configs. ✓/flag (P4)
- Does the formin pool cap mothers per node?** Yes — `dtsForminPool/Consume` give  $\sim 1$  mother/hot-vertex. ✓
- `forminRecover` = optional re-recruitment, not unlimited?** Yes, default 0 = finite. ✓
- `motherTetherDepth` biologically useful placement?** Works on the legacy path; the DTS analog `dtsForminStandoff` (0.05  $\mu$ m off-surface) is the validated one. ✓ (DTS); legacy param unused.
- `membraneAlignAngle` = barbed dir vs outward normal, correct?** Legacy mechanism is correct ( $90^\circ$ =tangent,  $<90^\circ$ =dendritic tilt) but **doesn't run on DTS.** (P2)
- `membraneAlignTorque` too rigid for floppy formin?** Opposite — on DTS it's *absent*; the DTS torque spring targets the wrong frame at  $\sim$ zero strength. (P2)
- `membraneCapDist` localizes growth without killing the network?** Mechanism sound but **disabled** (=0) on the live path. (P1)
- `branchMembraneDist` realistic / not too volumetric?** Legacy param unused; DTS uses a fixed  $\sim 60$  nm barbed-tip-near-cortex gate — appropriately thin, near the membrane. ✓
- `arpLocalField` plausible NPF field?** Legacy version superseded; DTS `dtsArp*` field is a correct source/diffuse/sink. ✓
- `membraneFaceCollideOn` makes a real surface?** That flag doesn't exist on DTS; `dtsActinCollide` drives a genuine continuous-surface collision. ✓ (caveat P3 for legacy drag).
- `membraneYield` converts push  $\rightarrow$  protrusion without instability?** DTS yields natively (no flag); the stability risk is the  $1/\sin\theta$  bending blow-up (P5), not the caps.
- Old params misleading/unused?** Yes — the entire legacy node-lattice param family is live-in-code, dead-on-path. Deprecate/document. (see §6)

5. Recommended parameter-only changes PARAMS

- 1. **Enable cortex-localized capping on dendritic DTS configs (P1).** Set `membraneCapDist ≈ 0.1–0.15 μm` in `dtsMembraneDendritic` / `dtsMembraneForminBranch` so barbed ends cap once they fall >~1 monomer behind the cortex. `end2TipC` already feeds it. *Highest-value biology fix; config-only.*
- 2. **Keep `dtsForminRecover:0`, `dtsForminPool:1`, `forminRelease:0`** for bonded studies — already correct in the bond/branch configs; don't raise without intent.
- 3. **Leave `membraneAlignTorque` low/off** as a deliberate floppy-formin choice — but see code change C3 (it does nothing on DTS regardless).
- 4. **`dtsBranchRate` : keep moderate (20–40).** Branch count  $\approx \text{dtsBranchRate} \cdot \text{arp}$  is exponentially sensitive — tune density here, length via `biochemDeltaT / actinConc`.

6. Recommended minimal code / documentation changes CODE / DOC

| ID | Change                                           | Detail                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|----|--------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| C1 | Model note (doc)                                 | Add to <code>Membrane.java</code> header and design §6: "Formin is implicit: it seeds linear mother filaments at hot Rac/Cdc42 (NPF) vertices and holds their barbed ends a standoff off the cortex. Arp2/3 branches daughters off those mothers; it never de-novo nucleates. Capping localizes growth to the leading edge; collision + ratchet convert polymerization into protrusion. FH1/FH2 mechanics are not resolved — this is a coarse mother-source/anchor." |
| C2 | Fix comment (doc)                                | Correct <code>Membrane.forminStep:535–536</code> (P6) — it describes the legacy pointed-end path; the active path holds the barbed end out.                                                                                                                                                                                                                                                                                                                          |
| C3 | Resolve <code>nodeTorqSpring</code> (small code) | Either retarget it to the true outward membrane normal (replace the body-frame <code>forminVecInx</code> target with the vertex outward radial) or set it to 0 and document that DTS mother orientation is seed-geometry-only (P2). Current state — "aligns to an arbitrary frame" — is neither. <i>Left for jba to choose direction.</i>                                                                                                                            |
| C4 | Deprecate legacy (doc)                           | Mark the legacy node-lattice membrane and its param family <code>@deprecated – superseded by buildDtsMembrane</code> in <code>Env.java</code> ; change <code>membraneNodeDragScale</code> default to 1.0 (P3) so the legacy path isn't a latent trap.                                                                                                                                                                                                                |
| C5 | Scope de-novo paths (doc)                        | Note that ActA / ProteinNode de-novo nucleation are non-dendritic (Listeria / test) and intentionally off in lamellipodium configs (P4).                                                                                                                                                                                                                                                                                                                             |
| C6 | Diagnostics (small code)                         | Extend the <code>[DTS-E]</code> log with: mothers-per-hot-vertex, branches-per-mother, mean local <code>arpLocal</code> over hot patches, barbed-tip → membrane distance histogram, mother & daughter angle-vs-normal distributions, peak radial displacement. Most inputs already exist.                                                                                                                                                                            |

7. Recommended validation run

Build on `dtsMembraneForminBranch` (already: one hot patch, `dtsForminPool:1`, `dtsArp0n`, `dtsActinCollide`, `forminRelease:0`):

- **Run A — capping/orientation baseline:** as-is, add `membraneCapDist:0.12`, stiff membrane (`dtsKappaArea:0.15`, `dtsTargetReducedVol:1.0` — "yield off"). Outputs: mother count, branch count, **capping fraction**, barbed-tip → membrane distance, protrusion displacement (expect ~0).
- **Run B — yield on:** same, soften the volume constraint (`dtsKappaVolume:1e-14`) so the cortex can bulge. Outputs: same metrics + radial displacement (expect a bounded protrusion at the patch).

Compare A vs B for: finite mothers ( $\approx 1/\text{hot-vertex}$ ), branches localized to the patch, capping fraction rising away from the cortex, and push → protrusion only in B. Watch `[DTS-E] |F|max` for the P5  $1/\sin\theta$  blow-up.

**Bottom line.** The live DTS lamellipodium module is in good biological shape on the hard requirements — formin-only mothers, Arp2/3-only branching off them, a real reaction-diffusion NPF field, a genuine continuous-surface collision, and a real Mogilner–Oster ratchet. The gaps are (1) cortex-localized **capping is implemented but switched off**, (2) mother **orientation bias toward the normal isn't wired on DTS** (the good mechanism is legacy-only; the DTS one points at the wrong frame), and (3) a large, mostly-dead **legacy parameter surface** that makes the model hard to read. None require a refactor — one config flag, one frame fix, and documentation close most of it.